

by construction

$Sv = 0$ and $lb \le v \le ub$ hold exactly at v^*, the ceilings with them

v^* is piecewise constant in the bounds and carries no gradient; only z^* does

the forced flux of the maintenance reaction (lb 0.7 mmol/gDW/h) must be carried inside the ceilings

measured (rounds 15 to 15d)

an infeasible program returns the zero flux and no gradient, so the state is absorbing (rounds 15b to 15d); duals on exchange bounds belong to no gene, so with no ceiling binding the backbone receives nothing (round 15b)